

CODE :-

```
import numpy as np
x1=np.array([1,1,1])
y1=np.array([1,1,-1,-1])
x2=np.array([-1,-1,1])
y2=np.array([-1,-1,1,1])

W1=np.outer(x1,y1)+np.outer(x2,y2)
print("w=",W1)

print("\n----- FORWARD CALL -----")
print("INPUT:")
print("original x1=",x1)
print("original x2=",x2)
print("\nOUTPUT:")
y1=np.sign(np.dot(x1,W1))
print("y1=",y1)
y2=np.sign(np.dot(x2,W1))
print("y2=",y2)

print("\n----- BACKWARD CALL -----")
print("INPUT:")
print("original y1=",y1)
print("original y2=",y2)
print("\nOUTPUT:")
x1=np.sign(np.dot(y1,W1.T))
print("x1=",x1)
x2=np.sign(np.dot(y2,W1.T))
print("x2=",x2)
```

OUTPUT :-

```
w= [[ 2  2 -2 -2]
     [ 2  2 -2 -2]
     [ 0  0  0  0]]

----- FORWARD CALL -----
INPUT:
original x1= [1 1 1]
original x2= [-1 -1  1]

OUTPUT:
y1= [ 1  1 -1 -1]
y2= [-1 -1  1  1]
```

----- BACKWARD CALL -----

INPUT:

original $y_1 = [1 \ 1 \ -1 \ -1]$

original $y_2 = [-1 \ -1 \ 1 \ 1]$

OUTPUT:

$x_1 = [1 \ 1 \ 0]$

$x_2 = [-1 \ -1 \ 0]$